MINDBURNER MIDI PRODUCTS

MALICE

Virtual Analogue Synthesiser Module



User Manual | Firmware v1.4

Hand-built in Belfast, Northern Ireland

Welcome

MALICE is a **virtual analogue MIDI synthesiser module**: it recreates the classic subtractive sound of analogue synths digitally, and it has no keyboard of its own. Connect a MIDI keyboard, sequencer, groovebox or DAW to its MIDI IN and it plays as a fat monosynth or a 6-voice polysynth. Each voice has two band-limited oscillators with eight wave shapes, a sub-oscillator, pulse width modulation, a noise source, X-Mod and Sync, drive and a resonant filter (12 dB, 24 dB, or the screaming Polivoks-style POLI modes), shaped by an envelope and an LFO that can lock to tempo. A two-slot mod matrix routes the mod wheel, velocity, aftertouch, LFO, envelope or key to eight sound and effect destinations. A stereo chorus with adjustable speed and a delay finish the sound, and a simple step sequencer with accent and slide is built in. Everything is synthesised in real time by firmware written from scratch for this instrument.

Connections

Socket	Details
USB-C power	5 V from a USB phone charger (recommended for the quietest sound) or a computer port, which can add noise. USB-C lead not supplied.
DC power	9 V DC (9 to 12 V accepted), 500 mA or more, 5.5 x 2.1 mm plug. Either polarity is fine. Use a good-quality regulated adaptor: cheap unregulated (heavy transformer) adaptors can cause a high-pitched whine.
MIDI IN	Standard 5-pin DIN. Plays notes and receives CC, Program Change, pitch bend, MIDI Clock, Start and Stop.
MIDI THRU	Standard 5-pin DIN. Passes on everything arriving at MIDI IN, so you can chain another synth or drum machine after MALICE.
Audio out	Stereo line level, for a mixer, audio interface or powered speakers. The output is hot: start with your mixer gain low. Headphones can be used for checking, but will sound quieter and lighter on bass.

Start-up

At power-up MALICE shows the Mindburner logo, then the MALICE face, whose eyes flash red, and opens on the last patch you used.

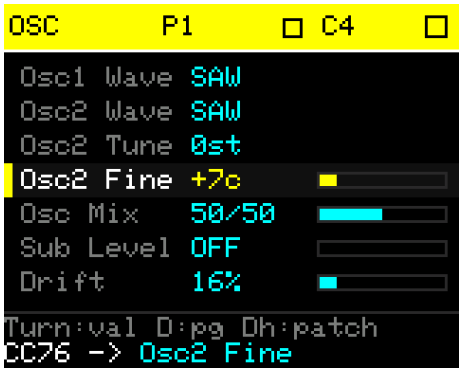
The controls

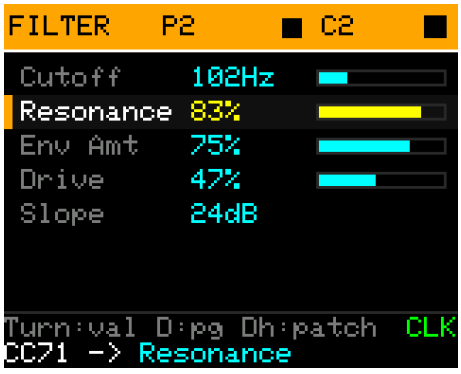
MALICE has four buttons, **A**, **B**, **C** and **D**, and a rotary encoder that also pushes as a button.

Sound pages

Control	What it does
Encoder turn	Change the selected value (turn quickly for bigger steps)
Encoder push	Next parameter. Hold: **previous** page
A tap / A hold	Next parameter / next page
B / C	Raise / lower the value (hold to repeat and speed up)
D tap / D hold	Next page / open the patch page
B+C hold	Open the patch page (and again to return)
A+B hold 1 s	Open the sequencer (and again to return)
A+B+C hold 2 s	Restore factory sounds (asks first: B = this slot, A = all 30, C = cancel)

*The header shows the page name, the patch number (with * if edited), MIDI activity, the last note played and whether a note is held. The footer names any CC as it arrives and shows messages. When MIDI clock is arriving, a green **CLK** appears bottom right, flashing white on each beat.*





The OSC page (left) and FILTER page (right). The selected row is highlighted, with a bar for continuous values.

Sound pages

OSC

Parameter	Description
Osc1 Wave / Osc2 Wave	SAW, TRI, SQR, PULSE, or the mixes SAW+SQR, SAW+TRI, TRI+SQR and SAW+PUL
Osc2 Tune	Plus or minus 12 semitones
Osc2 Fine	Detune 0 to +50 cents, for a thick, beating sound
Osc Mix	Balance between the two oscillators
Sub Level	Square wave one octave below oscillator 1
Drift	Gentle analogue-style pitch wander

PW/XMOD

Parameter	Description
Width	Pulse width, 50% (square) to 5% (thin)
PWM Depth	Amount of pulse width sweep
PWM Rate	Speed of the sweep, 0.05 to 20 Hz
X-Mod	Oscillator 2 modulates oscillator 1 (JX-8P / Prophet-style cross-mod): bells, metal and clang. OFF to 100%
Sync	OFF or ON: oscillator 1 hard-synced to oscillator 2. Sweep Osc2 Tune (or route ENV to OSC2 PITCH) for the classic tearing sync lead

X-Mod and Sync are clearest with Osc Mix towards oscillator 1. Tune osc 2 slightly off a musical interval with Osc2 Fine for metallic, inharmonic tones (patch 27 Metal Bell).

NOISE

Parameter	Description
Level	Noise level (OFF to full)
Type	WHITE, PINK, BROWN or DIGITAL
Colour	DARK, WARM or BRIGHT
Decay	HOLD (steady) or a burst from 5 ms to 2 s, for breath and chiff
Oscs	ON, or MUTED to hear the noise on its own (for percussion and effects)

FILTER

Parameter	Description
Cutoff	20 Hz to 20 kHz
Resonance	Emphasis at the cutoff
Env Amt	How far the envelope opens the filter
Drive	Saturation before the filter
Slope	12dB (smoother), 24dB (deeper, classic), POLI LP or POLI BP

POLI LP *and* ***POLI BP*** *are inspired by the Soviet Polivoks: the signal hits the filter hard and its stages saturate, so high Resonance screams and self-oscillates without running away. Drive pushes it further. Try patches 12 and 20.*

ENV

Parameter	Description
Attack	1 ms to 3 s
Decay	5 ms to 4 s
Sustain	Level while held
Release	5 ms to 5 s

One envelope shapes both the filter (through Env Amt) and the volume.

LFO

Parameter	Description
Wave	SINE, TRI, SAW, SQR or S&H (random steps)
Rate	0.05 to 20 Hz (shows SYNC when Rate Sync is on)
Rate Sync	OFF, 4 BAR, 2 BAR, 1 BAR, 1/2, 1/4, 1/8, 1/16, 1/4T or 1/8T: sets the speed from the tempo instead of Rate
To Pitch	Vibrato, up to plus or minus 12 semitones
To Filter	Filter sweep
To Amp	Tremolo
Sync	FREE or KEY (restarts with each new note)

MOD (mod matrix)

Parameter	Description
Source 1 / 2	OFF, WHEEL, VELOCITY, AFTERTCH (channel pressure), LFO, ENV or KEY (higher notes = more)
Dest 1 / 2	CUTOFF, PEAK, PITCH, OSC2 PIT, PULSE W, OSC MIX, LFO RATE, VOLUME, DLY MIX or CHO DPTH
Amount 1 / 2	-100% to +100%. Negative amounts work in reverse (e.g. the wheel closes the filter)
Wheel Vib	AUTO (vibrato from the mod wheel, switched off when WHEEL is used as a source), ON or OFF

Good starting points: WHEEL to CUTOFF +70%; VELOCITY to CUTOFF for brighter hard notes; AFTERTCH to PITCH for bends; WHEEL to DLY MIX +100% with Dly Mix at 0, so the echo fades in as you push the wheel (patch 13). For effects, VELOCITY, ENV and KEY follow the most recent note. Each patch stores its own matrix.

With Rate Sync on, each LFO cycle is started by MIDI clock when it is arriving, counted from MIDI Start, so a 1-bar sweep always begins on the downbeat. Without clock it follows the sequencer BPM. Set Rate Sync to OFF to return to the Rate setting.

FX

Parameter	Description
Chorus	OFF, I, II or I+II
Cho Rate	AUTO (classic I / II speeds) or 0.05 to 8 Hz. Slow settings also go deeper for a 70s swish
Delay	OFF, CLEAN, TAPE (warm, darkening repeats) or PING (ping-pong)
Time	20 to 580 ms (shows SYNC when Dly Sync is on)
Dly Sync	OFF, 1/4, 1/8D, 1/8, 1/8T or 1/16: locks the delay to the tempo
Feedback	Number of repeats
Dly Mix	Delay level

VOICE

Parameter	Description
Mode	MONO (last-note priority, legato) or POLY (6 voices)
Glide	Portamento, OFF to 2 s
Volume	Overall level
MIDI Ch	OMNI or 1 to 16 (global)
Tune A4	Master tune, 430 to 450 Hz (440 Hz standard). Global, not stored per patch
Random	B: a new random patch, C: undo the last one
Reboot	B, then B or push to confirm: restarts MALICE

Random builds a bass, lead, pad or pluck the way subtractive sounds are really made. It leaves Mode, Glide, Volume, the sync settings and the sequencer alone, so you can keep a pattern playing and press B to hear it on one new sound after another. Save the ones you like straight away, as undo goes back one step. Reboot saves the patch number, MIDI channel and sequencer edits before restarting.

Patch memory

Hold **B+C** (or hold **D**) to open the patch page: 30 slots, with the selected slot in yellow and the live patch in green. Turn the encoder or tap **A** to choose a slot. **Push the encoder to audition** it: MALICE switches to that sound and plays a short note, and your keyboard plays it too, so you can compare slots before loading. Press **B** to load or **C** to save, then confirm with B or push (yes) or C (no). Saving always stores your own sound, never the one you were auditioning, and leaving without loading puts your sound back as it was. Hold B+C or tap D to return. Program Change 0 to 29 also selects patches 1 to 30.

Restore factory sounds

Hold **A+B+C** for two seconds. A box asks what to restore: **B** (or push) restores the selected slot, **A** restores all 30, **C** cancels. Only patches are reset: the sequencer, MIDI channel and Tune A4 are kept.

Compare and recall

While auditioning, the header shows **PREVIEW**. **Hold the encoder** to go back to your own sound and hear it, so you can push and hold back and forth to compare. Hold the encoder after **loading** a patch and MALICE swaps back to the sound you had before, edits included; hold again to swap forward. The recovered sound shows as edited (for example P5*) until you save it. A good way to find a home for a new sound: push to hear what is in each slot, hold to get your sound back, then press C on the slot you choose.

Factory patches

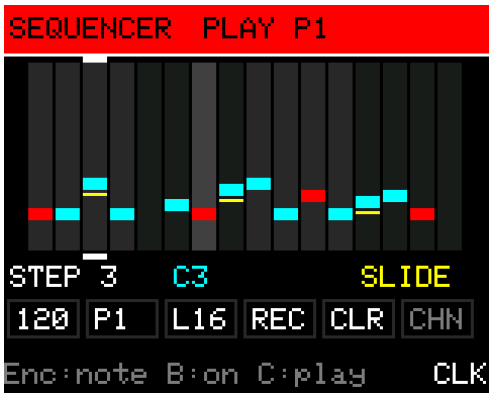
#	Patch	#	Patch
1	Classic	16	Warm Keys
2	Acid	17	Trance Lead
3	Vox Humana	18	Reese Bass
4	Pluck	19	Glass Pluck
5	Square Lead	20	Soviet Bass (POLI LP)
6	Sub Bass	21	Poli Brass
7	Wobble Bass	22	Acid Poli
8	S&H Bleeps	23	Talk Box
9	Strings	24	Dark Pad
10	Tremolo Keys	25	Echo Bleep
11	Brass	26	Cold Wind
12	Screamer Lead (POLI BP)	27	Metal Bell (X-Mod)
13	PWM Bass (wheel: tape echo)	28	Juno Bass
14	Evolving Pad	29	Ping-Pong Strings
15	Hoover	30	Glass Organ



The 30-slot patch page (left), and the MOD page with the wheel opening the filter and velocity setting the volume (right).

Simple step sequencer

MALICE includes a simple step sequencer for sketching basslines and riffs: 16 steps, 8 patterns, with acid-style accent and slide on every step. It is a handy extra rather than a full production sequencer, so it has no song mode, note lengths or per-step editing beyond note, accent and slide. Hold **A+B** for one second to open it, and again to return. It keeps playing in the background and saves automatically.



The grid is a mini piano roll: the height of each block is its pitch, cyan for a normal note, red for an accent, with a yellow line for a slide. White bars mark the selected step and a lighter column follows the playing step. Below are six boxes: **BPM**, **PAT**, **LEN**, **REC**, **CLR** and **CHN** (chain).

Sequencer controls

Control	What it does
A tap / A hold	Next / previous step or box
Encoder push	Jump between the step grid and the boxes (back to the same step)
Encoder turn	On a step: change its note. On a box: change its value
B tap / B hold	On a step: note on or rest / cycle accent and slide
D tap	On a step: cycle plain, accent, slide, both
D hold	Play / stop, from anywhere
C	Play / stop (on a step, REC or CLR); lower the value on BPM, PAT, LEN, CHN
B on boxes	Raise BPM, PAT, LEN or CHN; REC: arm recording; CLR: clear the pattern

When stopped, each note you set plays briefly so you can hear it.

Accent and slide

An **accent** plays louder and opens the filter further. A **slide** holds the note into the next step and glides to it without retriggering, even when Glide is off. Sliding into the same note gives a tie. Slides need a **MONO** patch; patch 2 **Acid** is ideal.

Patterns, chain and recording

P1 holds a starter acid riff and P2 a minor bassline; P3 to P8 are empty. Changing PAT while playing switches at the end of the pattern. LEN sets the length from 1 to 16 steps. Set **CHN** to, for example, >4 and MALICE plays P1 to P4 in turn, skipping empty patterns, with the grid following along. To record, arm **REC**, start the sequencer and play your keyboard: notes land on the nearest step, as accents if you play hard and as slides if you play legato.

Timing

The sequencer runs at its own BPM and automatically follows MIDI clock when it is arriving. MIDI Start begins exactly on the first clock beat and MIDI Stop stops it. With a chain set, MIDI Start always begins at the top of the chain (P1), so MALICE stays lined up with KLAK and other Mindburner sequencers.

MIDI implementation

Function	CC numbers
Oscillators	70 Osc1 wave, 80 Osc2 wave, 81 Osc2 tune, 76 Osc2 fine, 82 Osc mix, 78 Sub level, 83 Drift
Pulse / X-Mod	104 Width, 105 PWM depth, 106 PWM rate, 21 X-Mod, 20 Sync
Noise	111 Level, 112 Colour, 113 Decay, 114 Osc mute, 115 Type
Filter	74 Cutoff, 71 Resonance, 77 Env amount, 85 Drive, 86 Slope
Envelope	73 Attack, 75 Decay, 79 Sustain, 72 Release
LFO	87 Wave, 88 Rate, 117 Rate sync, 89 To pitch, 90 To filter, 102 To amp, 103 Sync
FX	93 Chorus, 94 Chorus rate, 107 Delay type, 108 Delay time, 116 Delay sync, 109 Feedback, 110 Delay mix
Performance	1 Mod wheel (vibrato), 5 Glide, 7 Volume, 120/123 All notes off, 126 Mono, 127 Poly

Also received: Note On/Off, velocity, channel aftertouch (mod matrix source), pitch bend (plus or minus 2 semitones), Program Change (patches 1 to 30), MIDI Clock, Start and Stop. The mod matrix settings have no CCs.

Tips and troubleshooting

If...	Try this
No sound	Check MIDI Ch on the VOICE page is OMNI or matches your controller, and that the mixer channel is up.
Slides overlap instead of gliding	Set Mode on the VOICE page to MONO.
The sequencer starts with my DAW	Your DAW sends MIDI Start. Stop MALICE with D hold, or leave the patterns empty.
The LFO or delay is out of time	Turn on Rate Sync (LFO page) or Dly Sync (FX page).
The chain skips patterns	Empty patterns are skipped. Add at least one note to include them.
A patch shows P3*	It has unsaved edits. Save it from the patch page, or reload to discard them.
Hum or zipper noise from USB	Power from a phone charger rather than a computer.
Whine on DC power	Try a good-quality regulated adaptor or USB power, and check the audio lead is pushed fully home.
Mod wheel adds vibrato I do not want	Set Wheel Vib on the MOD page to OFF (or AUTO with WHEEL used in the matrix).
A mod slot does nothing	Check its Amount is not 0%, and that the echo or chorus is switched on for DLY MIX and CHO DPTH.
Want the factory sounds back	Hold A+B+C for 2 s, then B for one slot or A for all.
Output too loud	The output is hot line level: lower the mixer gain or the Volume.

Mindburner MIDI Products, hand-built in Belfast, Northern Ireland.